Applied Labor Economics

Shortage Lists and Monopsony

Project Proposal

Lisa Chardon–Denizot, Maria Romagnoli

April 3rd, 2026

Context - *research question*

métiers en tension

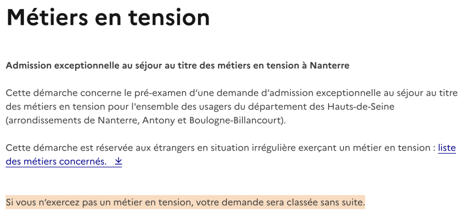
- Easier hiring of non-EEA workers in occupations facing labor shortages.
- List varies by region and occupation.
- Three updates (2008, 2021, 2025).

hypothesis

- Incentive for people seeking better status stability to work in shortage occupations.
- Employers know this → price wages down.

research question

Do non-EEA workers accept lower wages in shortage occupations because they give access to better legal status?



Métiers en tension

Admission exceptionnelle au séjour au titre des métiers en tension à Nanterre

Cette démarche concerne le pré-examen d'une demande d'admission exceptionnelle au séjour au titre des métiers en tension pour l'ensemble des usagers du département des Hauts-de-Seine (arrondissements de Nanterre, Antony et Boulogne-Billancourt).

Cette démarche est réservée aux étrangers en situation irrégulière exerçant un métier en tension : liste des métiers concernés.

Si vous n'exercez pas un métier en tension, votre demande sera classée sans suite.

Context - *motivation*

- Rich empirical research in this field suggests this could be the case.
  - **Amior & Stuhler (2023)** Germany after Iron Curtain. Immigration strengthened firms' monopsony power at the bottom of pay distribution. Large decline in wages.
  - **Signorelli (2024)** France 2008 shortage list reform. Increase in hiring of non-EU workers. Wages fell in shortage occupations more for natives than for migrants.
  - **Naidu, Nyarko & Wang (2016)** United Arab Emirates. Firms paid workers less than their marginal product. Reform untied workers from employers, wages rose.
- Our contribution: use more recent data than Signorelli (2024), try to explicitely isolate the *monopsony effect* from taste-based discrimination.

Context - *Alternative explainations*

- **Glover, Pallais & Pariente (2022, QJE)** — Managers can be biased about the productivity of migrant-worker performance, and this is a self-fulfilling prophecy, which can in turn rationalize lower wages in the mind of the firm
- More largely, any form of statistical discrimination
- It could be due to preferences of migrants to sort on firms that are lower paid but bring them some other utility gains (for instance many coworkers/a manager from the same origine).

How we address it: Compare non-EEA migrants with EEA-migrants that come from close countries.

Mechanism - *theoretical framework*

Several effects at play:

1. *permit/monopsony effect*: firms post lower wages because they have the bargaining power.
2. *shortage effect*: firms post higher wages to attract workers in shortage occupations.
3. *discrimination effect* firms offer lower wages to non-EEA workers because they're discriminating.

On monopsony effect:

- Could tie our reasoning to the classic job search model with Nash bargaining and outside option. Wage $= \underbrace{(1-\beta)\,\bar{u}_i}_{\text{outside option}} + \underbrace{\beta \cdot p_{ort}}_{\text{share of surplus}}$

Mechanism - *theoretical framework*

- There are three worker types: natives, EEA workers, and non-EEA workers. Their outside option is different !
  - natives and eea workers: outside option is market wage $\rightarrow \bar{w}_{\text{natives}} \approx \bar{w}_{\text{eea}} = \bar{w}^*$
  - non-eea workers: outside option is undocumented work, or new permit procedure $\rightarrow$ $\bar{w}_{noneea} = \max(\bar{w}^{\text{undoc}}, \underbrace{w^{\text{doc}} - \text{new permit cost}}_{\text{gain from being documented when there is an oppurtinity cost}}) \ll \bar{w}^*$
- This *monopsony effect* could explain the ethnic wage gaps.

$$\Delta w \equiv w_{eea} - w_{non-eea} = (1-\beta)(\bar{w}^* - \bar{w}_{non-eea}) > 0$$

More

The role of outside options

How is surplus shared? The key question is: which side (demand or supply) has the monopsony power?

- These occupations are characterized by labour supply rationing: less people want to work there, so firms need to push wages up, increasing the Nash bargaining power of migrants.
- However, firms know that non-EEA listed workers need to remain employed in a "métier en tensions"-occupation, and therefore enjoy a larger share of the bargaining power.

We use changes in the list and wage variations between EEA and non-EEA migrant as quasi-experimental variation to identify the monopsony power parameter, $\beta_1 * \beta_2$.

Empirics - *data*

ForCE

1. FH table → job-seeker records from *France Travail*.
   - socio-demographics.
   - reservation wage.
2. MMO table → private sector work contracts from *Déclaration Sociale Nominative*.
   - wage.
   - occupation code.

Shortage Lists

1. Digitalized *arrêtés* with FAP and ROME.

Empirics - *strategy*

Difference-in-differences :

$$\ln w_{i,o,r,t} = \alpha + \beta_1(\text{NonEEA}_i \times \text{Listed}_{o,r,t}) + \beta_2(\text{EEA}_i \times \text{Listed}_{o,r,t}) + \beta_3(\text{nonEEA}_i) + \beta_4(\text{nonEEA}_i) + \beta_5\text{Listed}_{o,r,t} + \gamma X_{it} + \phi_{ort} + \varepsilon_{i,o,r,t}$$

- ϕ_{ort}: occupation × region × year FE — absorbs demand-side wage effects of the listing and any cell-level shocks
- β_1: additional log wage gap for non-EEA workers *specifically* in listed occupations, relative to EEA workers in the *same* cell
- X_{it}: controld

Staggered difference-in-differences — Normalize the time-axis so that t=0 is the year during which $\text{Listed}_{o,r,t} = 1$ for the first time. **Event study** — plot $\hat{\delta}_k$ (differential wage trajectory of non-EEA vs. EEA) around the listing date to test pre-trends and trace the dynamic undercut.

Empirics - *Necessary tests*

- *Endogenous listing*: occupations may be listed *because* they already attract non-EEA workers at low wages $\Rightarrow$ test whether pre-listing wage gap predicts future listing;
- Parallel trend assumption

Conclusions - *our concerns*

1. Not managing to clean the data appropriately !
2. Statistical power issue

Thank you for listening!

Any questions ?

Appendix - *focusing on monopsony*

- How to separate the monopsony channel from taste-based discrimination ?
- If possible, compare groups similar on ethnic, cultural, religious lines.

Non-EEA (Count)	**EEA (Count)**	**Bias?**
Yugoslavs (34,000)	Croats + Slovenes (3,873)	Partial
Yugoslavs (34,000)	Bulgarians + Romanians (164,859)	Partial
Turks (111,043)	RO + BG + GR + CY (171,348)	Religious
Other Europeans (133,339)	Baltics + Central EU (61,839)	None

Note: Groups in the final row are Slavic (except Estonia and Hungary) and Christian. Need to assume no discrimination against Orthodoxy. Back